

Part 1

1. What goal (or goals) do you want your model to do? An example would be predicting the genre of a movie, or analyzing sentiment from a video. We want you to think about and discuss what is the end goal of the project that you will end up implementing later in the course.

I want to build a visuo-haptic material recognition model that predicts an object's material class from multimodal signals, with a focus on scenarios where vision is unreliable (e.g., occlusion, clutter, or ambiguous appearance).

2. List out any datasets that you can find that can help accomplish this. Explain why you think the data is relevant and in addition discuss any drawbacks of the dataset.

The main dataset I will use is the CLAMP dataset (<https://emprise.cs.cornell.edu/clamp/>).

I also identified related haptic datasets (e.g., TVL, Touch and Go), but CLAMP is the best fit for this project because it has larger scale and richer multimodal information for material recognition.

Why CLAMP is relevant:

- It is directly designed for material-related perception tasks.
- It contains multimodal signals that are useful for distinguishing materials with similar visual appearance.
- In addition to raw data, it provides processed data (.npz) that is practical for fast experimentation.

Drawbacks:

- Material classes are imbalanced, which may bias model training.
- Label noise and natural variability from in-the-wild collection may affect consistency.

3. What modalities do you choose to use? Why? Are there other modalities that could possibly be obtained that you don't plan on using? If so, why?

I use two modalities in this project: haptic time-series features and visual priors.

The haptic modality includes force, thermal, vibration, and proprioceptive-derived signals, which directly encode physical properties relevant to material identity.

The visual modality is represented by image-derived material probability vectors, which provide complementary appearance-based cues.

I chose these two modalities because they are complementary: haptics is strong for physical property discrimination, while visual priors help when appearance cues are reliable. Their combination is expected to be more robust than single-modality inputs.

Other modalities are possible, such as raw language/audio annotations and raw image/video streams. However, I do not prioritize them in this stage because my current objective is strictly material recognition with efficient multimodal feature integration; raw media can be added in later work for end-to-end representation learning.

4. What difficulties did you encounter in obtaining the data?

The major challenge was understanding the dataset packaging and deciding what to download first without over-downloading unnecessary files. The CLAMP dataset includes both raw archives (.zip) and processed files (.npz), so I needed to choose a practical entry point for reproducible experiments. After carefully reviewing the CLAMP paper and dataset metadata, I chose CLAMP_dataset_filtered.npz to ensure a stable and manageable workflow for this assignment.

Additionally, before selecting CLAMP, I surveyed other candidate touch/haptic datasets and found that some were not publicly accessible, some only released small subsets, and some were too large or fragmented to use efficiently within the homework timeline and local compute/storage constraints.

5. Recall the [six core challenges of multimodal learning](<https://arxiv.org/pdf/2209.03430>). How do you plan on addressing them in your dataset or anticipate each of them impacting the way you design your dataset?

1. **Representation:** I will learn modality-specific representations using haptic sequence features and visual prior vectors, and apply normalization to handle heterogeneity in scale and structure.

2. **Alignment:** I will align modalities at the interaction-instance level (haptic signals and visual priors from the same object-contact sample), which is a practical alignment strategy for material classification.
3. **Reasoning:** The reasoning objective is to infer latent physical properties (material identity) from complementary multimodal evidence, especially when one modality is ambiguous.
4. **Generation:** Generation is not the main focus of this project; however, I will test robustness under partial information by applying perturbation-style augmentations to available inputs (e.g., noise or degradation on visual priors / feature-level perturbations).
5. **Transference:** I will study transference by comparing unimodal and multimodal settings and evaluating whether representations learned under one condition generalize to other splits/subsets (e.g., different object subsets or class distributions).
6. **Quantification:** I will evaluate with Accuracy, Macro-F1, Balanced Accuracy, and MCC, and run ablations (haptic-only vs visual-prior-only vs multimodal) to quantify each modality's contribution and robustness.

Part 2

What difficulties did you encounter in extracting the modalities?

I encountered several practical challenges while extracting modalities from CLAMP.

First, the dataset release contains both raw archives (.zip) and processed files (.npz), so I needed to choose a source that supports a reproducible pipeline; I selected CLAMP_dataset_filtered.npz for this homework.

Second, the main dataset field is stored as an object array, which required explicit conversion into dense numeric tensors before any channel-wise processing.

Third, I had to use feature_key carefully to map semantic channel names to indices and avoid modality-splitting errors.

Finally, the visual priors are stored as per-sample dictionaries, so I converted them into fixed-length class-aligned vectors to make them usable in downstream multimodal analysis.

Overall, the main challenge was building a clean and reliable extraction workflow across heterogeneous data formats.

Part 3

What visualizations you tried, why, and submit what the visualizations looked like.

For this part, I used five visualizations to answer three goals:

(1) inspect class separability, (2) inspect sample-level signal behavior, and (3) inspect input statistics and channel relationships.

(Note that for visualization efficiency and readability, I used a reproducible random subset of 2,500 samples from the full dataset (fixed random seed). This sampling only affects plotting scope and does not change the modality extraction pipeline.)

1. **Visualizing Data Distribution:** t-SNE on multimodal summary features

Why: I used this to check whether material classes are naturally separable in a low-dimensional embedding before model training. If classes cluster clearly, classification should be easier; if they overlap heavily, stronger models/fusion are needed.



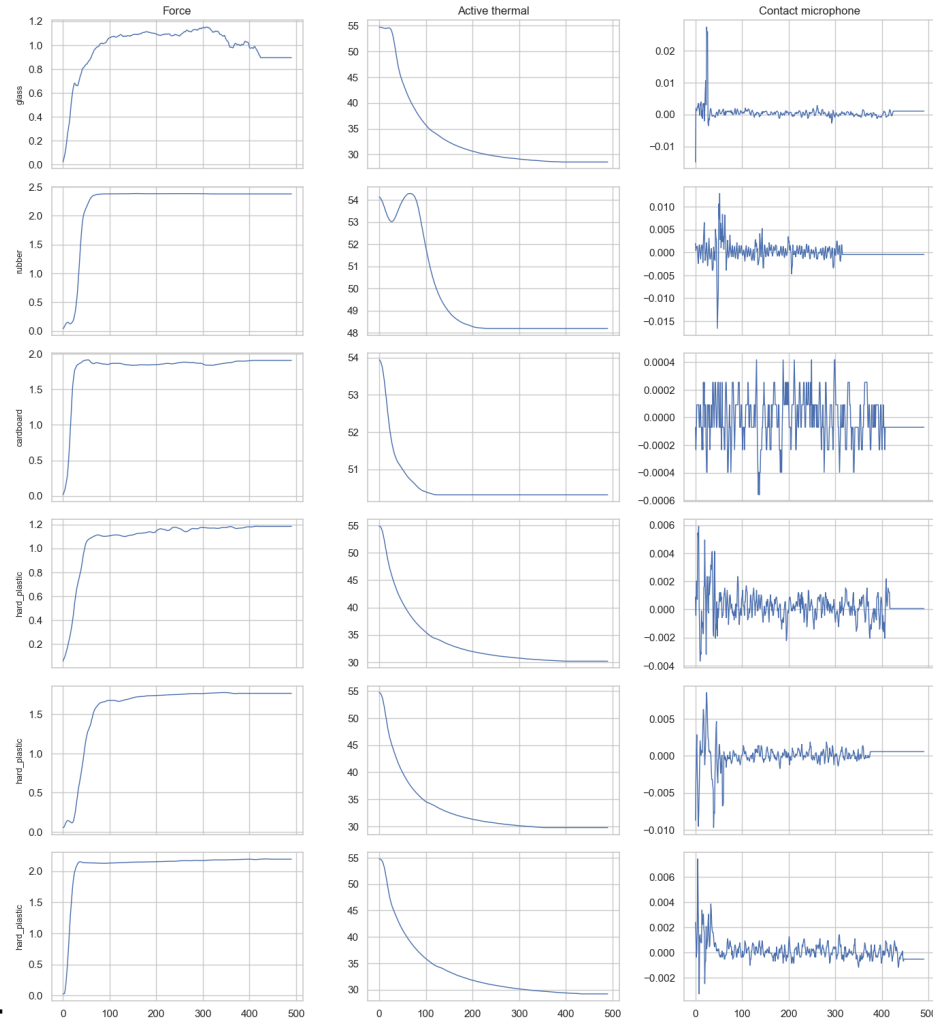
Result:

What it showed: The plot shows weak-to-moderate local grouping but substantial inter-class overlap across most regions. This indicates realistic class ambiguity and limited separability in the current feature space.

2. **Visualizing Samples:** multi-channel time-series examples

Why: I used sample-level trajectories to verify that the extracted modalities contain meaningful temporal patterns, not just noise. This is especially important for haptic sensing, where dynamics over time carry material information.

Sample-level Multimodal Signals

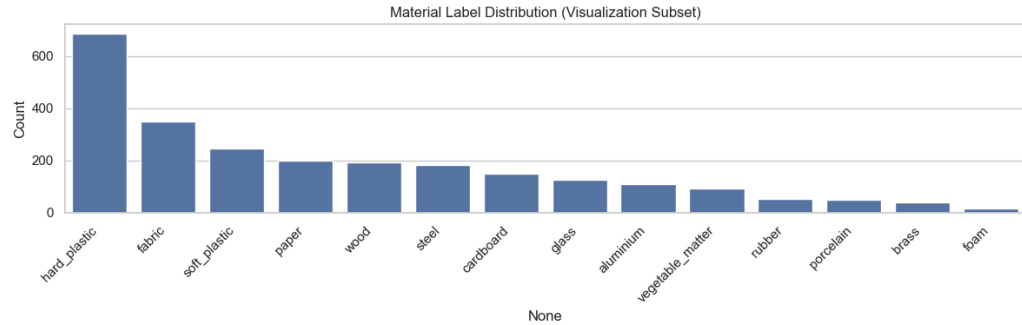


Result:

What it showed: Force curves generally rise and stabilize, active thermal curves decay over contact time, and contact microphone signals show transient bursts/noise-like variations. These distinct temporal behaviors support using multimodal temporal features for material recognition.

3. Visualizing Input Distribution (1): material label histogram

Why: I used this to diagnose label balance, which directly affects metric choice and training stability.

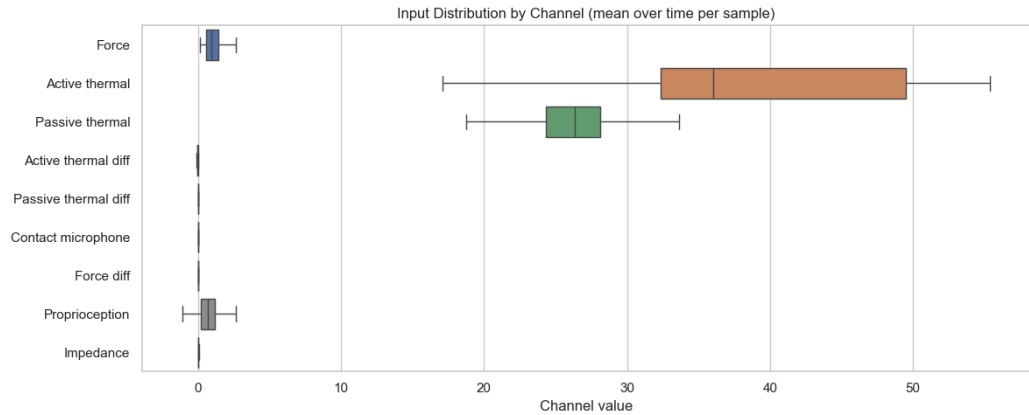


Result:

What it showed: The label distribution is clearly imbalanced (e.g., hard_plastic is much more frequent than foam/brass/porcelain). This justifies using imbalance-aware metrics such as Macro-F1, Balanced Accuracy, and MCC.

4. Visualizing Input Distribution (2): channel-level boxplots

Why: I used this to compare scale and spread across channels and identify preprocessing needs.

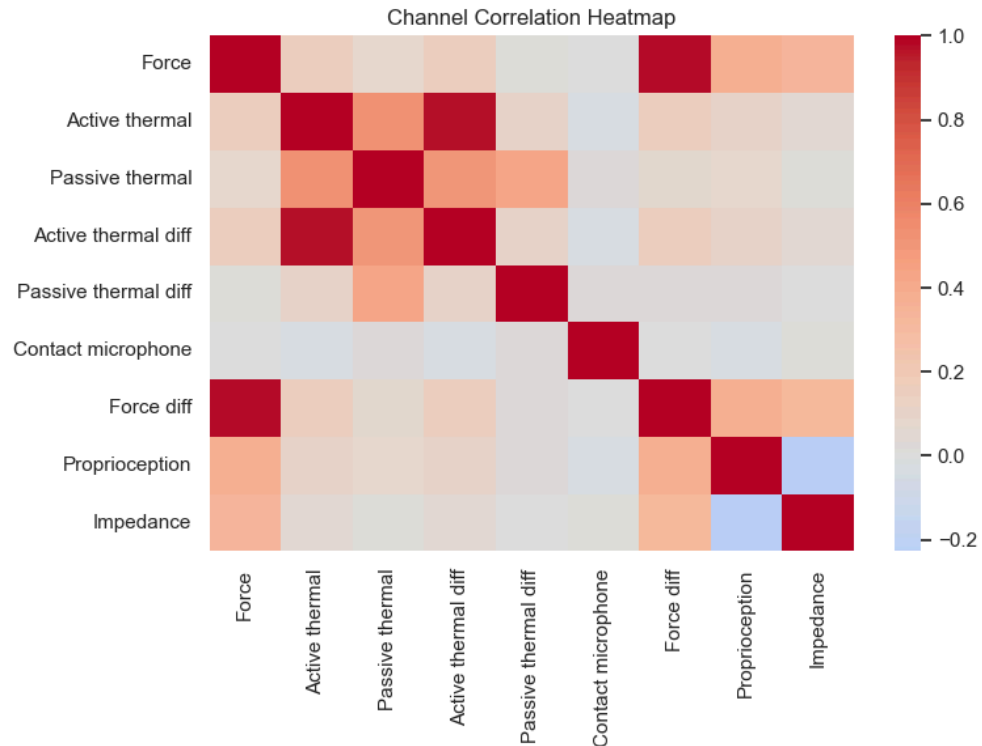


Result:

What it showed: Channels have very different value ranges (e.g., thermal channels dominate in magnitude while derivative/microphone channels are near zero). This confirms that normalization/standardization is necessary before fusion or distance-based analysis.

5. Additional exploratory visualization: channel correlation heatmap

Why: I used this to assess redundancy vs complementarity among channels and to motivate multimodal fusion design.



Result:

What it showed: Some channel pairs are strongly correlated (e.g., Force–Force diff, Active thermal–Active thermal diff), while others are weakly related or slightly negative (e.g., Proprioception–Impedance). This suggests both shared and unique information across modalities.

Overall, these visualizations provide both global and local evidence for my modeling choices: use multimodal fusion, normalize channel scales, and evaluate with imbalance-aware metrics.

Part 4

1. What evaluation metrics are you planning on using? Why?

I plan to use Accuracy, Macro-F1, Balanced Accuracy, and MCC (Matthews Correlation Coefficient).

- Accuracy gives an intuitive overall correctness score.
- Macro-F1 treats each class equally and is important because material classes are imbalanced.

- Balanced Accuracy averages recall across classes, making it less sensitive to dominant classes.
- MCC provides a robust single-score summary for multiclass classification under imbalance and prediction bias.

Using this combination gives a more complete view than relying on one metric alone.

2. Are there any other metrics that could be used here or that you considered?

Yes. I also considered:

- Weighted-F1, which reflects class frequency and is useful for deployment-oriented performance.
- Per-class precision/recall, to inspect which materials are systematically confused.
- Top-k accuracy (e.g., top-2), which can be useful when close material categories overlap.
- Calibration metrics (e.g., ECE/Brier score) for confidence-aware decisions.
- Confusion matrix-based analysis, for detailed error structure.

For this homework stage, I prioritized the four core metrics above because they are concise, interpretable, and strong for imbalanced multiclass evaluation.

3. List out the pros and cons of the evaluation metrics you decided to go with.

Accuracy

- Pros: Simple, easy to interpret, widely used.
- Cons: Can be misleading under class imbalance.

Macro-F1

- Pros: Gives equal importance to all classes; highlights minority-class performance.
- Cons: Can look pessimistic when some rare classes are very hard.

Balanced Accuracy

- Pros: Better class-balance sensitivity than plain accuracy.
- Cons: Does not capture precision-related errors directly.

MCC

- Pros: Robust summary metric under imbalance; uses all confusion information.
- Cons: Less intuitive for non-technical readers and harder to explain than accuracy.

Overall, this metric set balances interpretability (accuracy) and robustness (Macro-F1 / Balanced Accuracy / MCC).

Part 5

Scenario 1 (Restaurant review sentiment)

Prompt:

You are a sentiment classifier.

Classify the following review into exactly one label from this set: {positive, negative, neutral}.

Return only one lowercase word and nothing else.

Review: "This place stinks, the service was awful and the food was not cooked. I will never come back here!"

****Expected output format example:****

negative

Scenario 2 (Emotion from face image dataset: angry/sad/happy)

Prompt:

You are an emotion classification model.

Given one face image, choose exactly one label from this set: {angry, sad, happy}.

Rules:

- 1) Output must be exactly one lowercase word.
- 2) Do not include punctuation, confidence, or explanation.
- 3) If uncertain, choose the single best label from the allowed set.

****Expected output format example:****

angry

Scenario 3 (Information extraction from novel paragraph)

Prompt:

Extract information from the paragraph below.

Return output in strict JSON format with exactly these keys:

{"subject_name": "...", "destination": "...", "planned_activity": "..."}
Do not add extra keys or text.

Paragraph: "The man, Edgar, flew to Italy to hike the Alps. He was looking forward to going skiing there."

****Expected output format:****

{"subject_name": "Edgar", "destination": "Italy", "planned_activity": "skiing"}

Part 6

1. The most interesting topic discussed in this homework assignment.

The most interesting part of this homework for me was the visualization section. It gave me a chance to learn and survey different visualization methods and, more importantly, understand what each visualization is actually useful for. This part helped me connect data exploration with concrete modeling decisions instead of treating plots as just presentation figures.

2. A challenging aspect that you did not expect to deal with and what insights you used to address it.

A challenging aspect I did not expect was data accessibility in practice.

Many papers describe their datasets as “open,” but in reality some are partially released or hard to access. I also found that some truly open datasets are so large or fragmented that downloading and organizing them becomes a project by itself.

The key insight I used was to treat dataset selection as an engineering decision, not just a research preference. Instead of optimizing for maximum scale, I prioritized reproducibility and feasibility: choose a stable entry point, build a working pipeline first, and then expand to a larger scale only if needed. This staged strategy reduced risk and let me make steady progress.

3. How do you feel about the overall quality of your dataset? Is there anything lacking? What is particularly great about it?

Overall, I think the dataset quality is strong for material recognition because it is multimodal, relatively large, and collected in realistic conditions. A limitation is class imbalance. The strongest aspect is the diversity of haptic signals and the practical usefulness of the dataset for robust multimodal material classification.